

10.06 Creating Trigonometric Models

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
Trigonometric Functions And Models	Identify the angle measure, quadrant, or graph feature before selecting a formula.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

A. Practice: Creating Trigonometric Models

1.

$$\sin 30^\circ = \frac{1}{2}$$

2.

$$135^\circ \cdot \frac{\pi}{180}$$

3.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

4.

$$\sin 30^\circ = \frac{1}{2}$$

5.

$$135^{\circ} \cdot \frac{\pi}{180}$$

6.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

7.

$$\sin 30^{\circ} = \frac{1}{2}$$

8.

$$135^{\circ} \cdot \frac{\pi}{180}$$

9.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

10.

$$\sin 30^{\circ} = \frac{1}{2}$$

11.

$$135^{\circ} \cdot \frac{\pi}{180}$$

12.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

13.

$$\sin 30^{\circ} = \frac{1}{2}$$

14.

$$135^{\circ} \cdot \frac{\pi}{180}$$

15.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

16.

$$\sin 30^{\circ} = \frac{1}{2}$$

17.

$$135^{\circ} \cdot \frac{\pi}{180}$$

18.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

19.

$$\sin 30^{\circ} = \frac{1}{2}$$

20.

$$135^{\circ} \cdot \frac{\pi}{180}$$

B. Apply the key formulas

1.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

2.

$$B = \frac{2\pi}{T}$$

3.

$$y = A\sin(Bc) + D$$

4.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

5.

$$B = \frac{2\pi}{T}$$

6.

$$y = A\sin(Ba) + D$$

7.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

8.

$$B = \frac{2\pi}{T}$$

9.

$$y = A\sin(Bd) + D$$

10.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

11.

$$B = \frac{2\pi}{T}$$

12.

$$y = A\sin(Bb) + D$$

13.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

14.

$$B = \frac{2\pi}{T}$$

15.

$$y = A\sin(Be) + D$$

16.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

17.

$$B = \frac{2\pi}{T}$$

18.

$$y = A\sin(Bc) + D$$

19.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

20.

$$B = \frac{2\pi}{T}$$